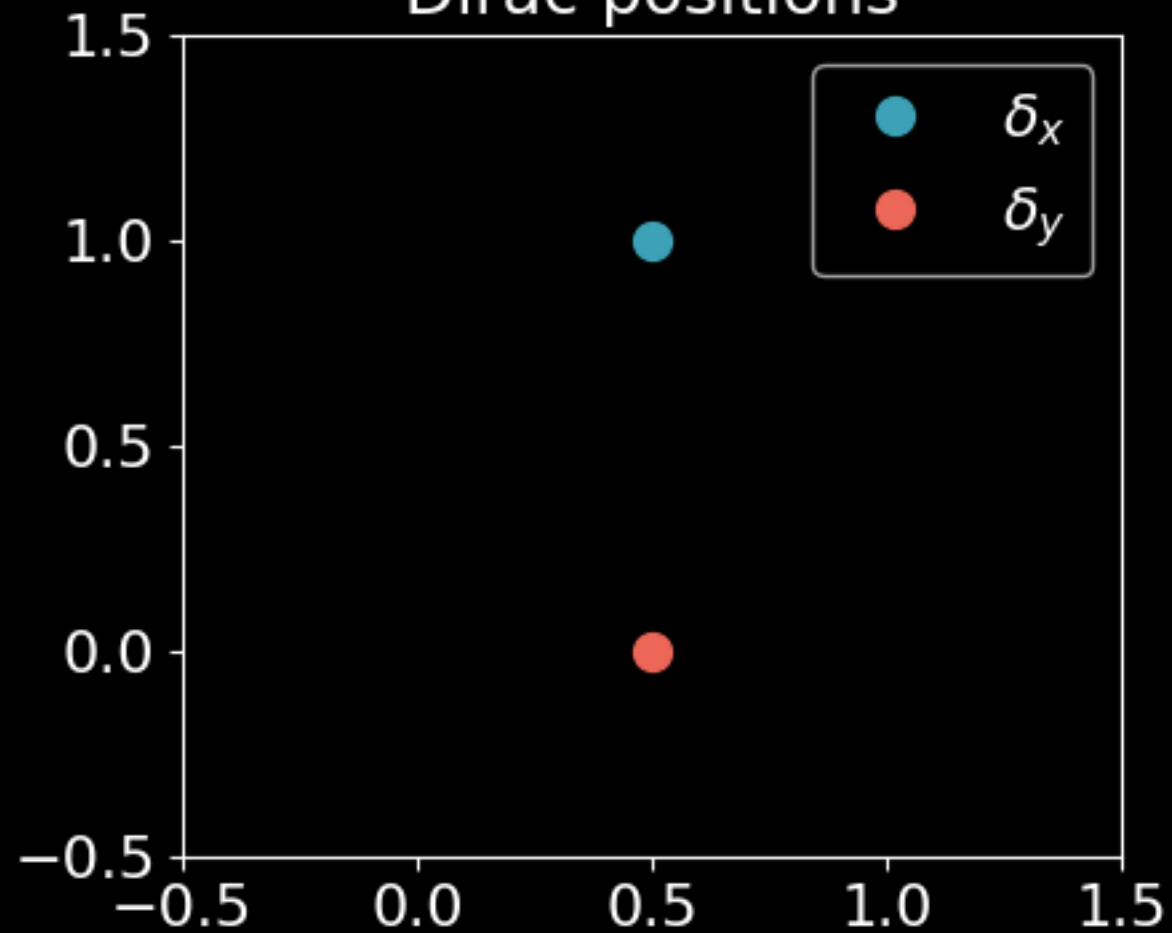
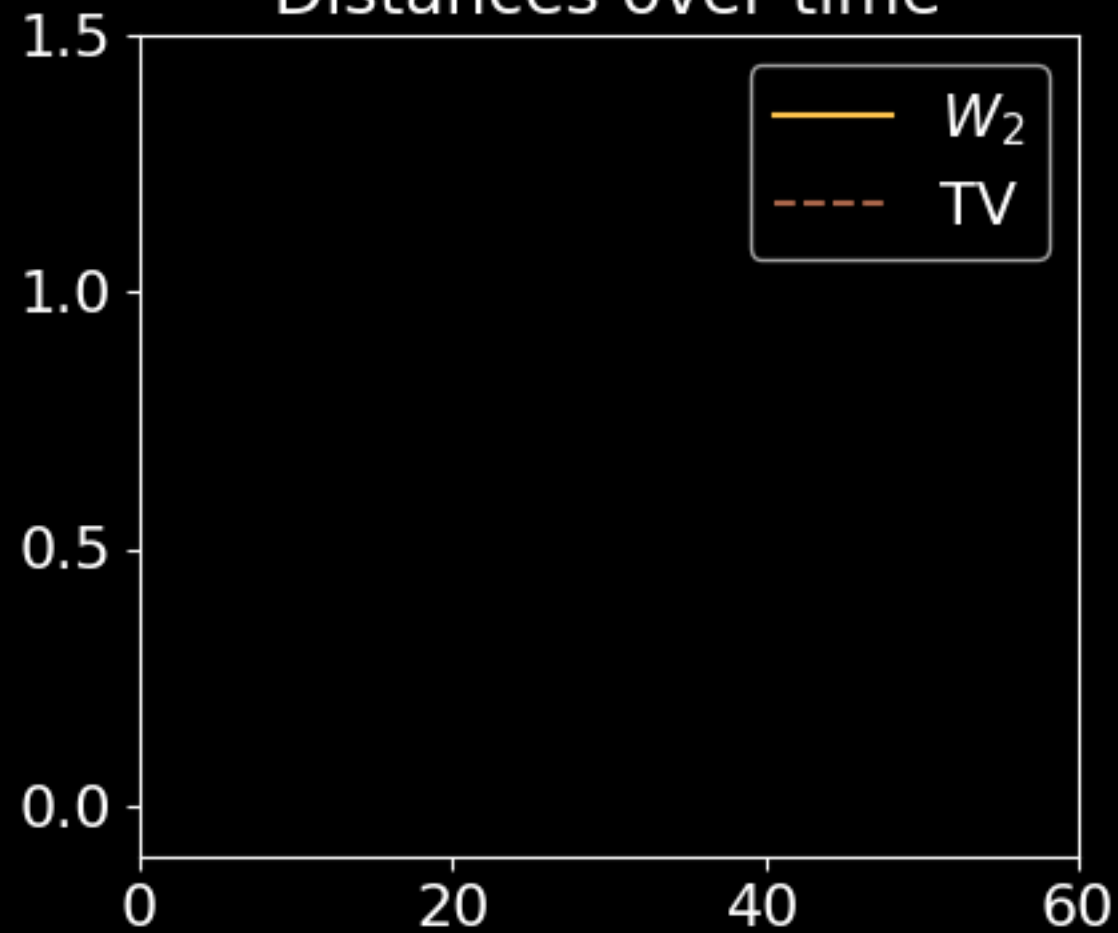


Dirac positions



Distances over time

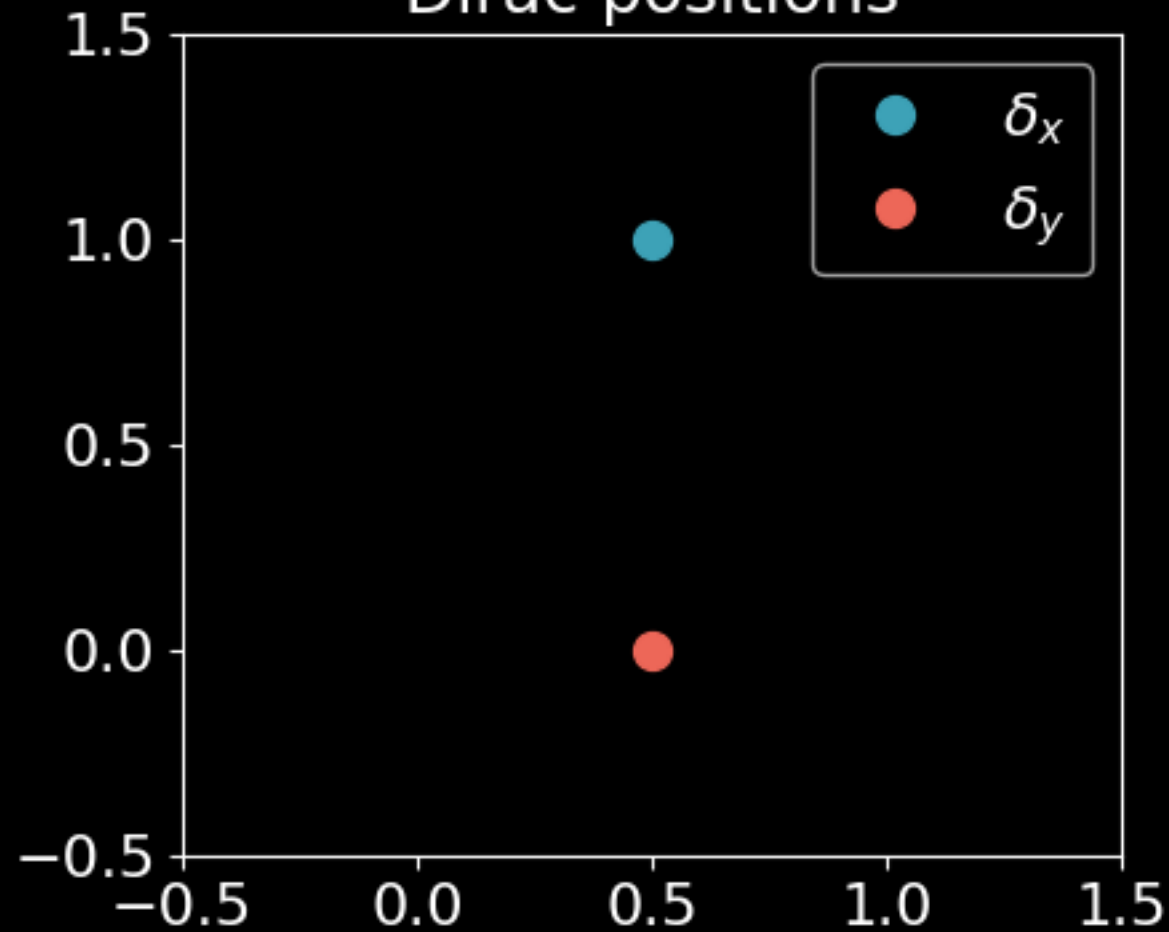




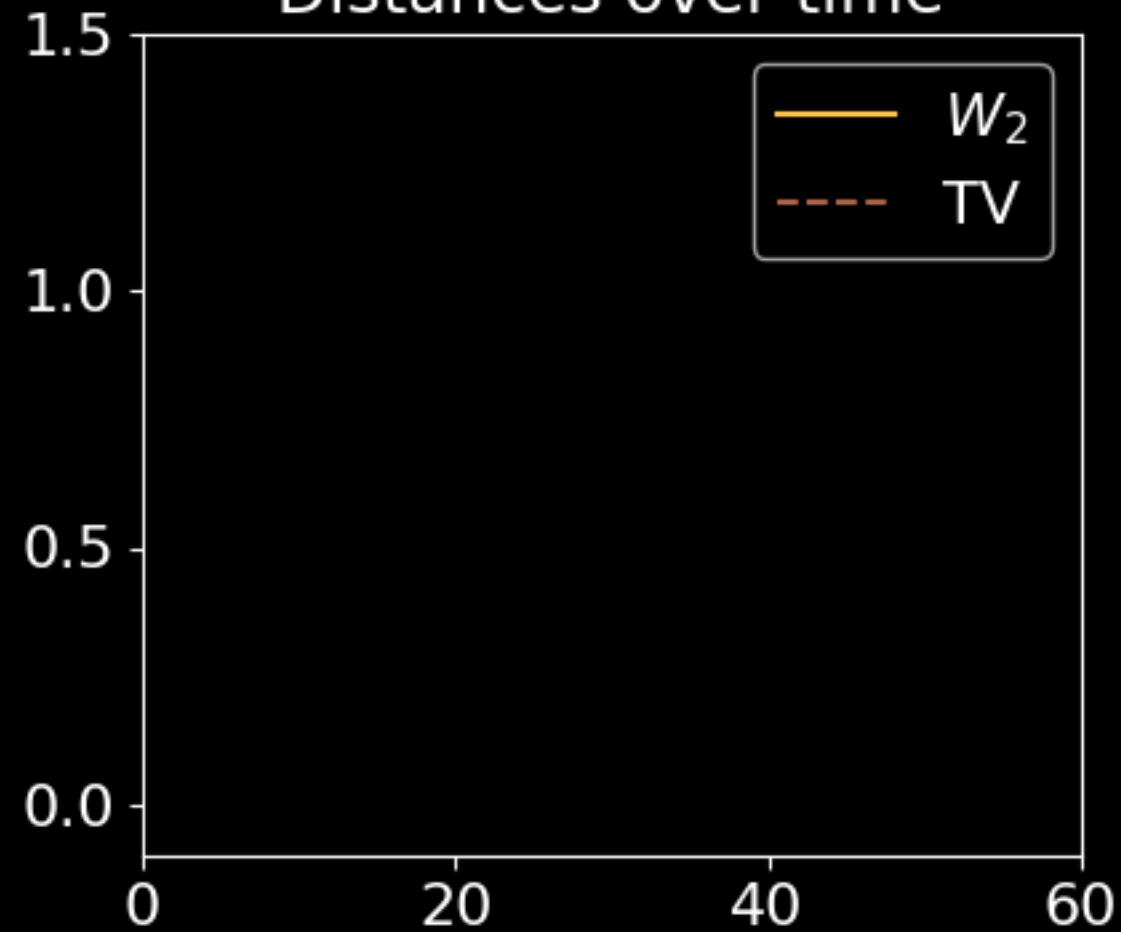




Dirac positions



Distances over time

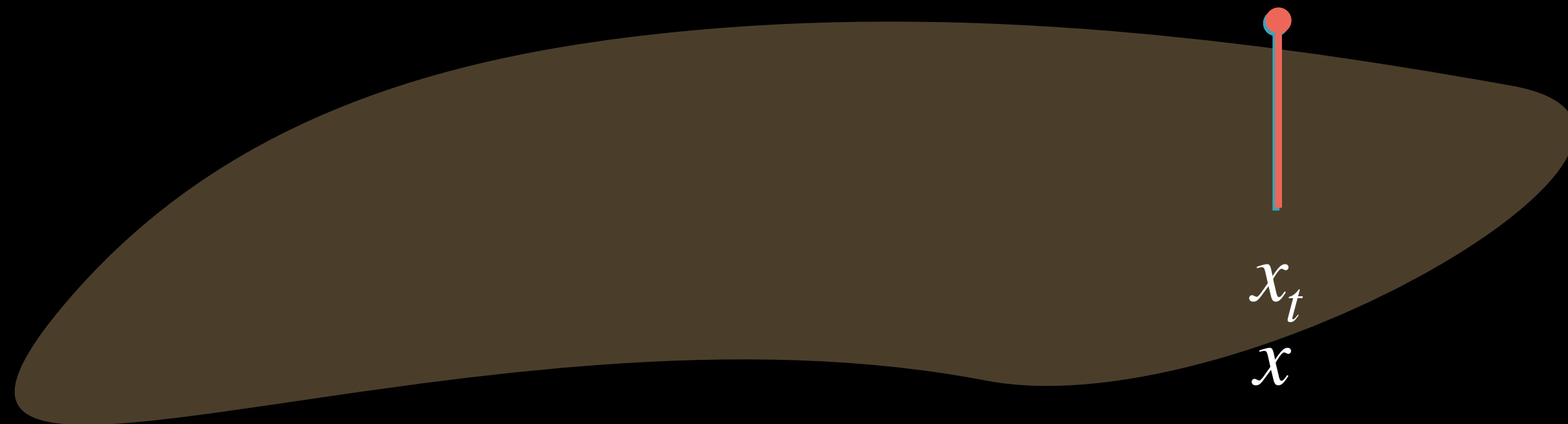


Why geometry awareness matters

$$\text{TV}(\delta_{x_t}, \delta_x) = 1$$

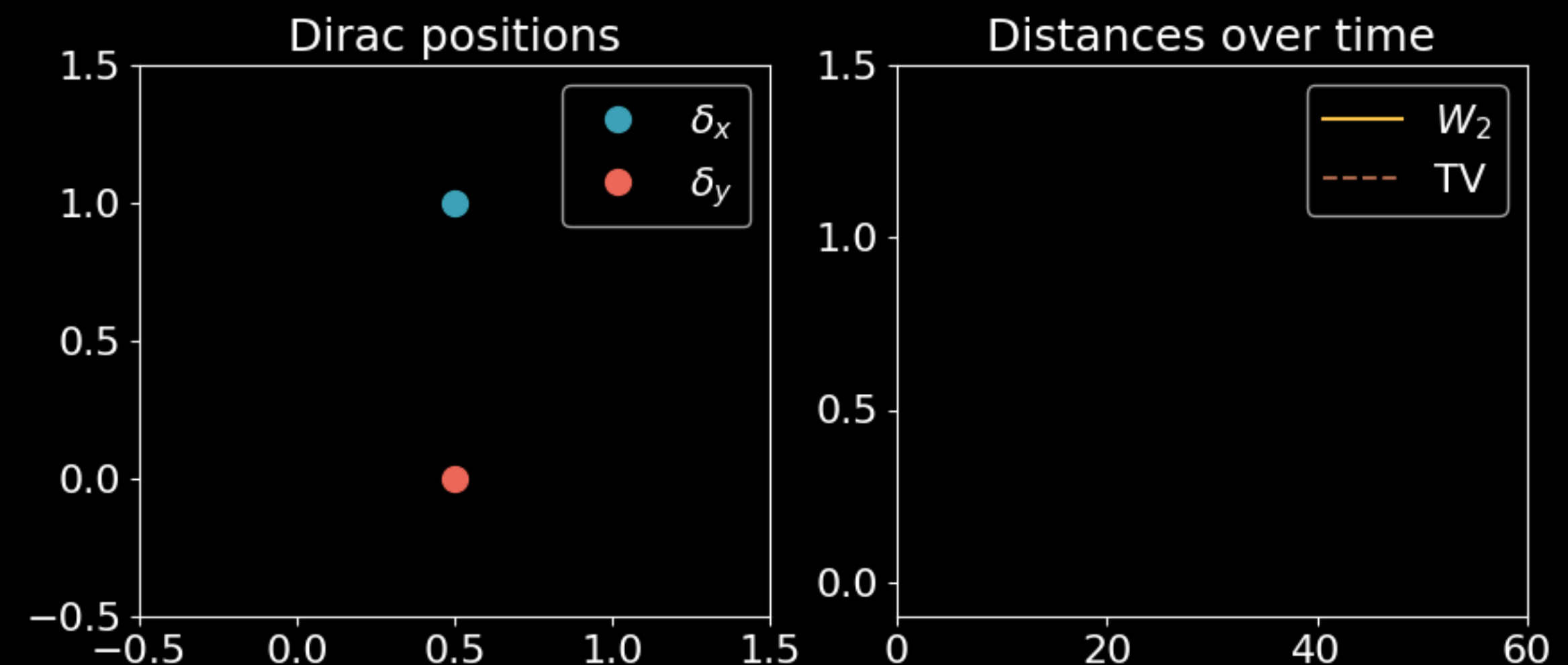
$$\text{KL}(\delta_{x_t}, \delta_x) = \infty$$

$$W_p(\delta_{x_t}, \delta_x) = d(x_t, x)$$



More generally:

- KL undefined if supports don't overlap
- TV don't consider geometry
- OT/Wasserstein respects underlying space!



KL, TV, JS vs. OT

	KL	TV	JS	Wasserstein
Symmetry	✗	✓	✓	✓
Finite if supports disjoint?	✗ (∞ if no overlap)	✓	✓	✓
Metric?	✗	✓	✗ (square root is a metric)	✓
Sensitivity	Very strong (blows up)	“All-or-nothing” differences	Softer, bounded in $[0, \sqrt{\log 2}]$	Smooth, reflects geometry
Interpretation	Relative entropy (information loss)	Max probability mismatch	Average divergence to midpoint distribution	Optimal transport cost
ML usage	Variational inference, ELBO	Hypothesis testing	GAN loss (original)	WGAN, domain adaptation, barycenters